QF 604
ECONOMETRICS OF FINANCIAL MARKETS

LECTURE 8

Professor Emeritus Lim Kian Guan

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LECTURE OUTLINE

01 COPULA

02 PROPERTIES OF COPULAS

03 TYPES OF COPULAS

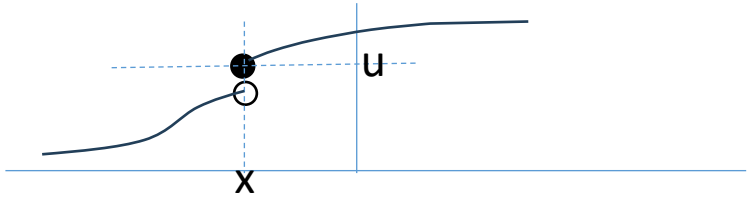
04 MEASURES OF DEPENDENCE

05 EXTREME VALUE COPULAS

06 ESTIMATING COPULAS

01 Preliminaries

- Standard Uniform CDF is G(U = u) = u for $0 \le u \le 1$. PDF is g(u) = 1 for $0 \le u \le 1$. A random variable X with CDF $F(X \le x) = u$ occurs with same probability as a standard uniform r.v. with CDF $G(U \le u)$. CDF is right continuous, i.e., $\lim_{y \downarrow x} F(y) = F(x)$, i.e., it is defined for $F(X \le x)$. U is also called the generalized inverse for CDF F.



- Continuous bivariate probability

$$P(a < X < b \quad \text{and} \quad c < Y < d) = \int_c^d \int_a^b f(x,y)dxdy$$

Given a bivariate probability density f(x, y), then we can calculate the marginal probability densities of X and of Y :

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy \quad \text{for all} \quad x, \qquad f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx \quad \text{for all} \quad y.$$

- Bivariate normal density function f(x), x ~ 2 x 1 vector

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \qquad \boldsymbol{\mu} = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix} \qquad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}$$

$$f(\mathbf{x}) = \frac{1}{2\pi}(\det \boldsymbol{\Sigma})^{-1/2} \exp\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right]$$

01 Copula

A copula or a copula function, as introduced by Sklar (1959) , is a joint distribution function of standard uniform variates. Suppose the underlying random variables are Z_i for $i = 1, 2, \ldots, n$. Each Z_i has a marginal probability distribution function (or cumulative distribution function CDF) $F(z_i) = u_i$ where Z_i takes realized value z_i. If Z_i is continuous in the range $(-\infty, +\infty)$, and $F(\cdot)$ is continuous and strictly increasing, then u_i is the realized value of a random variable U_i with a standard uniform distribution in the range $(0, 1)$.

We can denote a n-dimensional copula function by $C(u_1, u_2, \ldots, u_n) = Pr(U_1 \leq u_1, U_2 \leq u_2, \ldots, U_n \leq u_n)$. For each u_i, we can find a value z_i such that $F^{-1}(u_i) = z_i$. Then $C(u_1, u_2, \ldots, u_n) = Pr(Z_1 \leq z_1, Z_2 \leq z_2, \ldots, Z_n \leq z_n)$. We denote the last term on the right-hand side as the joint probability distribution function $P(z_1, z_2, \ldots, z_n)$. Therefore, a copula function $C(F(z_1), F(z_2), \ldots, F(z_n)) = P(z_1, z_2, \ldots, z_n)$ and it characterizes the joint probability distribution function of the underlying random variables $Z_1, Z_2, \ldots, Z_n$. Just as $Z_1, Z_2, \ldots, Z_n$ may or may not be correlated, similarly $U_1, U_2, \ldots, U_n$ may or may not be correlated.

01 Copula

More formally:

Definition 1 *A d-dimensional copula, $C : [0,1]^d \; : \rightarrow \; [0,1]$ is a cumulative distribution function (CDF) with uniform marginals.*

We write $C(\mathbf{u}) = C(u_1, \ldots, u_d)$ for a generic copula and immediately have (why?) the following properties:

1. $C(u_1, \ldots, u_d)$ is non-decreasing in each component, u_i.

   CDF is (weakly) increasing in each u_i, keeping other components constant.

2. The ith marginal distribution is obtained by setting $u_j = 1$ for $j \neq i$ and since it it is uniformly distributed

$$C(1, \ldots, 1, u_i, 1, \ldots, 1) \;=\; u_i.$$

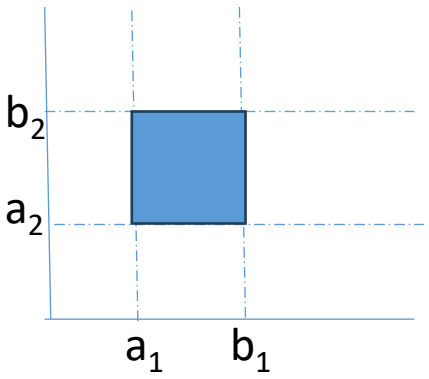
   $\int_0^1 C(u_1, u_2) du_2 = C(u_1, 1) = u_1$

3. For $a_i \leq b_i$, $P(U_1 \in [a_1, b_1], \ldots, U_d \in [a_d, b_d])$ must be non-negative. This implies the rectangle inequality

$$\sum_{i_1=1}^{2} \cdots \sum_{i_d=1}^{2} (-1)^{i_1 + \cdots + i_d} \, C(u_{1,i_1}, \ldots, u_{d,i_d}) \;\geq\; 0$$

where $u_{j,1} = a_j$ and $u_{j,2} = b_j$.

For example in $C(u_1, u_2)$, for $a_i \leq u_i \leq b_i$ in i =1,2, $[C(b_1,b_2) - C(b_1,a_2)] - [C(a_1,b_2) - C(a_1,a_2)]$, any volume in d-dimension cube is (weakly) positive



(2) And (3) imply that $C(u_1,\ldots,u_{j-1},0,u_{j+1},\ldots.) = 0$ if $u_j = 0$ for any j. In the above example, set $a_1 = b_1 = 0$, and see that $C(0, u_2) = 0$.

01 Copula

The generalized inverse, $F^{\leftarrow}$, is defined as

$$F^{\leftarrow}(x) \;:=\; \inf\{v \;:\; F(v) \geq x\}$$

Proposition 1 *If $U \sim U[0,1]$ and F_X is a CDF, then*

$$P\left(F^{\leftarrow}(U) \leq x\right) \;=\; F_X(x).$$

In the opposite direction, if X has a continuous CDF, F_X, then

$$F_X(X) \;\sim\; U[0,1].$$

Now let $\mathbf{X} = (X_1, \ldots, X_d)$ be a multivariate random vector with CDF F_X and with continuous and increasing marginals. Then by Proposition 1 it follows that the joint distribution of $F_{X_1}(X_1), \ldots, F_{X_d}(X_d)$ is a copula, C_X say. We can find an expression for C_X by noting that

$$\begin{aligned} C_X(u_1, \ldots, u_d) &= P\left(F_{X_1}(X_1) \leq u_1, \ldots, F_{X_d}(X_d) \leq u_d\right) \\ &= P\left(X_1 \leq F_{X_1}^{-1}(u_1), \ldots, X_d \leq F_{X_d}^{-1}(u_d)\right) \\ &= F_{\mathbf{X}}\left(F_{X_1}^{-1}(u_1), \ldots, F_{X_d}^{-1}(u_d)\right). \end{aligned} \tag{1}$$

If we now let $u_j := F_{X_j}(x_j)$ then (1) yields

$$F_{\mathbf{X}}(x_1, \ldots, x_d) = C_X(F_{X_1}(x_1), \ldots, F_{X_d}(x_d))$$

This is one side of the famous Sklar's Theorem which we now state formally.

01 Copula

Theorem 2 (Sklar's Theorem 1959) *Consider a d-dimensional CDF, F, with marginals $F_1, \ldots, F_d$. Then there exists a copula, C, such that*

$$F(x_1, \ldots, x_d) \;=\; C\left(F_1(x_1), \ldots, F_d(x_d)\right) \tag{2}$$

for all $x_i \in [-\infty, \infty]$ and $i = 1, \ldots, d$.

If F_i is continuous for all $i = 1, \ldots, d$, then C is unique; otherwise C is uniquely determined only on $\mathrm{Ran}(F_1) \times \cdots \times \mathrm{Ran}(F_d)$ where $\mathrm{Ran}(F_i)$ denotes the range of the CDF, F_i.

In the opposite direction, consider a copula, C, and univariate CDF's, $F_1, \ldots, F_d$. Then F as defined in (2) is a multivariate CDF with marginals $F_1, \ldots, F_d$. $\square$

01 Copula

Example 1 Let Y and Z be two IID random variables each with CDF, $F(\cdot)$. Let $X_1 := \min(Y, Z)$ and $X_2 := \max(Y, Z)$ with marginals $F_1(\cdot)$ and $F_2(\cdot)$, respectively. We then have

$$P(X_1 \leq x_1,\ X_2 \leq x_2) \ = \ 2\, F\left(\min\{x_1, x_2\}\right)\ F(x_2) \ - \ F\left(\min\{x_1, x_2\}\right)^2. \tag{3}$$

We can derive (3) by considering separately the two cases (i) $x_2 \leq x_1$ and (ii) $x_2 > x_1$.

We would like to compute the copula, $C(u_1, u_2)$, of (X_1, X_2). Towards this end we first note the two marginals satisfy

$$\begin{aligned} F_1(x) &= 2F(x) - F(x)^2 \\ F_2(x) &= F(x)^2. \end{aligned}$$

show

show

But Sklar's Theorem states that $C(\cdot, \cdot)$ satisfies $C(F_1(x_1),\ F_2(x_2)) \ = \ F(x_1, x_2)$ so if we connect the pieces(!) we will obtain

$$C(u_1, u_2) \ = \ 2\ \min\{1 - \sqrt{1 - u_1},\ \sqrt{u_2}\}\sqrt{u_2} \ - \ \min\{1 - \sqrt{1 - u_1},\ \sqrt{u_2}\}^2.$$

show

02 Properties of Copula

when marginals are continuous

The Density and Conditional Distribution of a Copula

If the copula has a density, i.e. a PDF, then it is obtained in the usual manner as

$$c(\mathbf{u}) = \frac{\partial^d C(u_1, \ldots, u_d)}{\partial u_1 \cdots \partial u_d}.$$

$$c(\mathbf{u}) = \frac{f\left(F_1^{-1}(u_1), \ldots, F_d^{-1}(u_d)\right)}{f_1\left(F_1^{-1}(u_1)\right) \cdots f_d\left(F_d^{-1}(u_d)\right)}$$

where we have used $F_i^{\leftarrow} = F_i^{-1}$ since F_i is differentiable. It is also easy to obtain the conditional distribution of a copula. In particular, we have

$$\begin{aligned} P\left(U_2 \leq u_2 \mid U_1 = u_1\right) &= \lim_{\delta \to 0} \frac{P\left(U_2 \leq u_2,\ U_1 \in (u_1 - \delta, u_1 + \delta]\right)}{P\left(U_1 \in (u_1 - \delta, u_1 + \delta]\right)} \\ &= \lim_{\delta \to 0} \frac{C(u_1 + \delta, u_2) \ - \ C(u_1 - \delta, u_2)}{2\delta} \\ &= \frac{\partial}{\partial u_1} C(u_1, u_2). \end{aligned}$$

This implies that the conditional CDF may be derived directly from the copula itself.

02 Properties of Copula

Proposition 3 (Invariance Under Monotonic Transformations) *Suppose the random variables $X_1, \ldots, X_d$ have continuous marginals and copula, C_X. Let $T_i : \mathbb{R} \to \mathbb{R}$, for $i = 1, \ldots, d$ be strictly increasing functions. Then the dependence structure of the random variables*

$$Y_1 := T_1(X_1), \ \ldots, \ Y_d := T_d(X_d)$$

is also given by the copula C_X.

Sketch of proof when T_j's are continuous and $F_{X_j}^{-1}$'s exist:
First note that

$$\begin{aligned} F_Y(y_1, \ldots, y_d) &= P(T_1(X_1) \le y_1, \ \ldots \ , T_d(X_d) \le y_d) \\ &= P(X_1 \le T_1^{-1}(y_1), \ \ldots \ , X_d \le T_d^{-1}(y_d)) \\ &= F_X(T_1^{-1}(y_1), \ \ldots \ , T_d^{-1}(y_d)) \end{aligned} \tag{6}$$

so that (why?) $F_{Y_j}(y_j) = F_{X_j}(T_j^{-1}(y_j))$. This in turn implies

$$F_{Y_j}^{-1}(y_j) = T_j(F_{X_j}^{-1}(y_j)). \tag{7}$$

The proof now follows because

$$\begin{aligned} C_Y(u_1, \ldots, u_d) &= F_Y\left(F_{Y_1}^{-1}(u_1), \ \ldots \ , F_{Y_d}^{-1}(u_d)\right) \text{ by (5)} \\ &= F_X\left(T_1^{-1}\left(F_{Y_1}^{-1}(u_1)\right), \ \ldots \ , T_d^{-1}\left(F_{Y_d}^{-1}(u_d)\right)\right) \text{ by (6)} \\ &= F_X\left(F_{X_1}^{-1}(u_1), \ \ldots \ , F_{X_d}^{-1}(u_d)\right) \text{ by (7)} \\ &= C_X(u_1, \ldots, u_d) \end{aligned}$$

and so $C_X = C_Y$. $\square$

02 Properties of Copula

Theorem 4 (The Fréchet-Hoeffding Bounds) *Consider a copula* $C(\mathbf{u}) = C(u_1, \ldots, u_d)$. *Then*

$$\max\left\{1 - d + \sum_{i=1}^{d} u_i,\ 0\right\} \ \leq\ C(\mathbf{u})\ \leq\ \min\{u_1, \ldots, u_d\}.$$

Sketch of Proof: The first inequality follows from the observation

$$\begin{aligned} C(\mathbf{u}) &= P\left(\bigcap_{1\leq i\leq d}\{U_i \leq u_i\}\right) \\ &= 1 - P\left(\bigcup_{1\leq i\leq d}\{U_i > u_i\}\right) \\ &\geq 1 - \sum_{i=1}^{d} P(U_i > u_i) = 1 - d + \sum_{i=1}^{d} u_i. \end{aligned}$$

The second inequality follows since $\bigcap_{1\leq i\leq d}\{U_i \leq u_i\} \subseteq \{U_i \leq u_i\}$ for all i. $\square$

02 Properties of Copula

Proposition 5 *Let $X_1, \ldots, X_d$ be random variables with continuous marginals and suppose $X_i = T_i(X_1)$ for $i = 2, \ldots, d$ where $T_2, \ldots, T_d$ are strictly increasing transformations. Then $X_1, \ldots, X_d$ have the comonotonic copula.*

Proof: Apply the *invariance under monotonic transformations* proposition and observe that the copula of $(X_1, X_1, \ldots, X_1)$ is the comonotonic copula. □

The **countermonotonic** copula is the 2-dimensional copula that is the Fréchet-Hoeffding lower bound. It satisfies

$$W(u_1, u_2) \;=\; \max\{u_1 + u_2 - 1,\ 0\} \tag{8}$$

and corresponds to the case of *perfect negative dependence*.

02 Properties of Copula

- For a bivariate distribution, show how a comonotonic copula corresponds to a case of perfect positive dependence, i.e., for any X, Y = cX where c is a positive constant.

- For a bivariate distribution, show how a counter-monotonic copula corresponds to a case of perfect negative dependence, i.e., for any X, Y = – cX where c is a positive constant.

- Show the joint distribution of (U, 1 - U) where U $\sim$ U(0, 1).

- Why is the Frechet-Hoeding lower bound not a copula for d > 2?

03 Types of Copula

Consider the bivariate standard normal density function of two random variables x and y:

$$\frac{1}{2\pi\sqrt{1-\rho^2}}\exp\left(\frac{-x^2-y^2+2\rho xy}{2(1-\rho^2)}\right).$$

The marginal densities of x and y are $(2\pi)^{-1/2}\exp(-\frac{1}{2}x^2)$ and $(2\pi)^{-1/2}\exp(-\frac{1}{2}y^2)$ respectively. Using Eq. (18.1), we obtain the Gaussian copula density as

$$c\left(\Phi(x),\Phi(y)\right)=\frac{1}{\sqrt{1-\rho^2}}\exp\left(\frac{x^2+y^2}{2}-\frac{x^2+y^2-2\rho xy}{2(1-\rho^2)}\right),$$

where $\Phi(\cdot)$ denotes the standard normal distribution function.

In the above case, using the definition of a copula function as $C(F(z_1), F(z_2),\ldots, F(z_n)) = P(z_1, z_2, \ldots, z_n)$, we have $C(\Phi(x),\phi(y)) = P(X \leq x, Y \leq y)$ which is

$$\int_{-\infty}^{x}\int_{-\infty}^{y}\frac{1}{2\pi\sqrt{1-\rho^2}}\exp\left(\frac{-v^2-w^2+2\rho vw}{2(1-\rho^2)}\right)dv\,dw\,.$$

$$c(F(z_1),F(z_2),\ldots,F(z_n))=\frac{f(z_1,z_2,\ldots,z_n)}{f_1(z_1)f_2(z_2)\ldots f_n(z_n)}\,, \qquad (18.1)$$

03 Types of Copula

In the above case, using the definition of a copula function as $C(F(z_1), F(z_2), \ldots, F(z_n)) = P(z_1, z_2, \ldots, z_n)$, we have $C(\Phi(x), \phi(y)) = P(X \leq x, Y \leq y)$ which is

$$\int_{-\infty}^{x} \int_{-\infty}^{y} \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(\frac{-v^2 - w^2 + 2\rho vw}{2(1-\rho^2)}\right) dv\, dw\,.$$

This is the Gaussian copula function. More formally, it can be written as

$$C(h,k) = \int_{-\infty}^{\Phi^{-1}(h)} \int_{-\infty}^{\Phi^{-1}(k)} \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(\frac{-v^2 - w^2 + 2\rho vw}{2(1-\rho^2)}\right) dv\, dw\,.$$

In a similar way, the Student t-distribution copula or t-copula $C_t(u_1, u_2, \ldots, u_n)$ can be evaluated as:

$$\int_{-\infty}^{t_v^{-1}(u_1)} \cdots \int_{-\infty}^{t_v^{-1}(u_n)} \frac{\Gamma\left((v+n)/2\right)}{\Gamma(v/2)\sqrt{(\pi v)^n |\Sigma|}} \left(1 + \frac{x' \Sigma^{-1} x}{v}\right)^{-(v+n)/2} dx,$$

where t_v is a standard univariate student-t distribution with v degrees of freedom, Σ is the correlation matrix of the $t_v^{-1}(u_i)$ variables implied by the standard uniform $u_i \in (0,1)$'s, and dx denotes a vector format.

04 Measures of Dependence

- Pearson, i.e. linear, correlation coefficient is only defined if second moments exist. It is invariant under positive linear transformations, but not under general strictly increasing transformations.

  Example: Cov (a + b X, Y) = Cov (X, Y)

- Rank correlations only depend on the unique copula of the joint distribution and are invariant to strictly increasing transformations. This is due to the Proposition 3 of copula invariance under monotonic transformations.

- Coefficients of tail dependence are a measure of dependence in the extremes of the distributions.

Theorem 6 *Let (X_1, X_2) be a random vector with finite-variance marginal CDF's F_1 and F_2, respectively, and an unspecified joint CDF. Assuming $\mathrm{Var}(X_1) > 0$ and $\mathrm{Var}(X_2) > 0$, then the following statements hold:*

1. *The attainable correlations form a closed interval $[\rho_{min}, \rho_{max}]$ with $\rho_{min} < 0 < \rho_{max}$.*
2. *The minimum correlation $\rho = \rho_{min}$ is attained if and only if X_1 and X_2 are countermonotonic. The maximum correlation $\rho = \rho_{max}$ is attained if and only if X_1 and X_2 are comonotonic.*
3. *$\rho_{min} = -1$ if and only if X_1 and $-X_2$ are of the same type. $\rho_{max} = 1$ if and only if X_1 and X_2 are of the same type.*

For given univariate distributions, F_1 and F_2, and any correlation value $\rho \in [-1, 1]$, it is not always possible to construct a joint distribution F with margins F_1 and F_2 and correlation ρ .

04 Measures of Dependence

There are two important rank correlation measures, namely Spearman's rho and Kendall's tau.

Definition 3 *For random variables X_1 and X_2,* **Spearman's rho** *is defined as*

$$\rho_s(X_1, X_2) \;:=\; \rho(F_1(X_1), F_2(X_2)).$$

Spearman's rho is therefore simply the linear correlation of the probability-transformed random variables. The Spearman's rho *matrix* is simply the matrix of pairwise Spearman's rho correlations, $\rho(F_i(X_i), F_j(X_j))$. It is (why?) a positive-definite matrix. If X_1 and X_2 have continuous marginals then it can be shown that

$$\rho_s(X_1, X_2) \;=\; 12 \int_0^1 \int_0^1 \left(C(u_1, u_2) \;-\; u_1 u_2\right) \, du_1 \, du_2.$$

It is also possible to show that for a bivariate Gaussian copula

$$\rho_s(X_1, X_2) \;= \frac{6}{\pi} \arcsin \frac{\rho}{2} \;\simeq\; \rho$$

where ρ is the Pearson, i.e., linear, correlation coefficient.

See proof in "Rank Correlation and Product-Moment Correlation" by P. A. P. Moran, Biometrika Vol. 35, No. 1/2 (May, 1948), pp. 203-206 (4 pages)

04 Measures of Dependence

Definition 4 *For random variables X_1 and X_2,* **Kendall's tau** *is defined as*

$$\rho_\tau(X_1, X_2) \;:=\; \mathrm{E}\left[\mathit{sign}\left((X_1 - \tilde{X}_1)\,(X_2 - \tilde{X}_2)\right)\right]$$

where $(\tilde{X}_1, \tilde{X}_2)$ is independent of (X_1, X_2) but has the same joint distribution as (X_1, X_2).

Note that Kendall's tau can be written as

$$\rho_\tau(X_1, X_2) \;=\; P\left((X_1 - \tilde{X}_1)\,(X_2 - \tilde{X}_2) \;> 0\right) \;-\; P\left((X_1 - \tilde{X}_1)\,(X_2 - \tilde{X}_2) \;< 0\right)$$

so if both probabilities are equal then $\rho_\tau(X_1, X_2) = 0$. If X_1 and X_2 have continuous marginals then it can be shown that

$$\rho_\tau(X_1, X_2) \;=\; 4 \int_0^1 \int_0^1 C(u_1, u_2)\, dC(u_1, u_2) \;-\; 1.$$

For a bivariate Gaussian copula,

$$\rho_\tau(X_1, X_2) \;= \frac{2}{\pi} \arcsin \rho$$

show

where ρ = Pearson correlation coefficient.

04 Measures of Dependence

Properties of Spearman's Rho and Kendall's Tau

Spearman's rho and Kendall's tau are examples of rank correlations in that, when the marginals are continuous, they depend only on the bivariate copula and not on the marginals. They are therefore invariant in this case under strictly increasing transformations.

Spearman's rho and Kendall's tau both take values in $[-1, 1]$:

- They equal 0 for independent random variables. (It's possible, however, for dependent variables to also have a rank correlation of 0.)
- They take the value 1 when X_1 and X_2 are comonotonic.
- They take the value -1 when X_1 and X_2 are countermonotonic.

04 Measures of Dependence

Tail Dependence

We have the following definitions of lower and upper tail dependence.

Definition 5 *Let X_1 and X_2 denote two random variables with CDF's F_1 and F_2, respectively. Then the* **coefficient of upper tail dependence**, *λ_u, is given by*

$$\lambda_u := \lim_{q \nearrow 1} P\left(X_2 > F_2^{\leftarrow}(q) \mid X_1 > F_1^{\leftarrow}(q)\right)$$

provided that the limit exists. Similarly, the **coefficient of lower tail dependence**, *λ_l, is given by*

$$\lambda_l := \lim_{q \searrow 0} P\left(X_2 \leq F_2^{\leftarrow}(q) \mid X_1 \leq F_1^{\leftarrow}(q)\right)$$

provided again that the limit exists.

If $\lambda_u > 0$, then we say that X_1 and X_2 have *upper tail dependence* while if $\lambda_u = 0$ we say they are *asymptotically independent in the upper tail. Lower tail dependence* and *asymptotically independent in the lower tail* are similarly defined using λ_l. The upper and lower coefficients are identical for both the Gaussian and t copulas. Unless $\rho = \pm 1$ it can be shown that the Gaussian copula is asymptotically independent in both tails.

04 Measures of Dependence

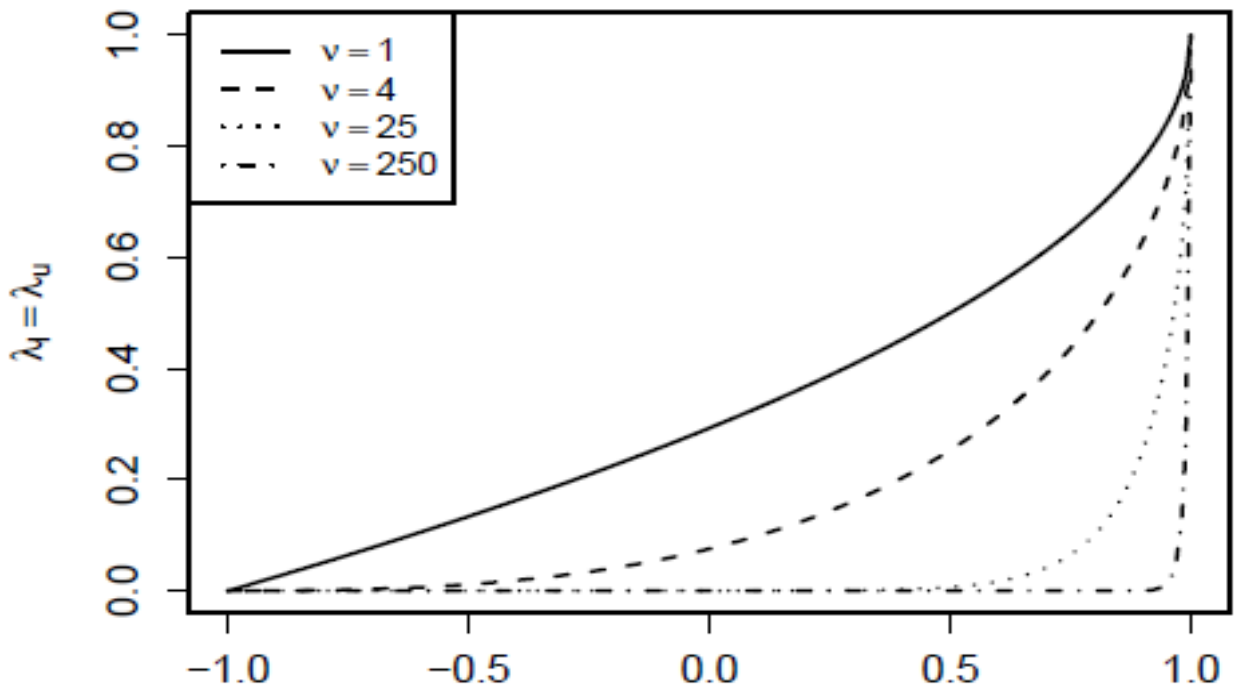


Figure 8.6 from Ruppert and Matteson: Coefficients of tail dependence for bivariate t-copulas as functions of for = 1, 4, 25 and 250.

04 Measures of Dependence

Choosing the right correlation or comparing them – Pearson ρ or Spearman's ρ_S or Kendall's tau ρ_τ

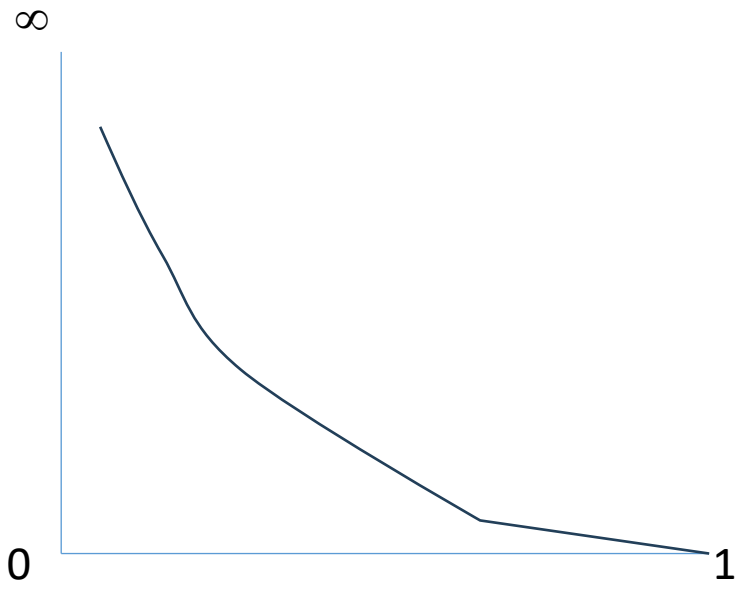
- ρ good if data X, Y are linearly related and normal. Captures all necessary information. But it is sensitive to both magnitude and direction of linear associations.
- ρ suitable for continuous data, not for ordinal or categorical data.
- ρ_S is non-parametric measure. Does not assume a linear relationship between X, Y. But assumes a monotonic relationship, i.e., how strong is the monotonic relationship $r(X_1) \leq r(Y_1)$, $r(X_2) \leq r(Y_2)$, ….. where r(.) is some kind of ranking, e.g., CDF.
- ρ_S is suitable for both continuous and ordinal data. It is robust to outliers.
- ρ_τ is non-parametric measure and makes no assumption about the data distribution. Does not assume a linear relationship between X, Y. It is suitable for both continuous and ordinal data. It is robust to outliers.
- ρ_τ measures monotonic relationship. Like ρ_S , it focuses on direction and orderings of data. If $(X_i - X_j)(Y_i - Y_j) > 0$, pairs (X_i,Y_i) , (X_j,Y_j) , are concordant. Otherwise discordant. ρ_τ is explicit measure of the probability of observing concordant (agreeable) vs. discordant (disagreeable) pairs. May not be reliable for small sample size.

05 Copulas with Large Tail Dependency

Family of Archimedean Copulas

Archimedean copulas C take the form:

$$C(u_1, \ldots, u_n) = \phi^{-1}\big(\phi(u_1) + \cdots + \phi(u_n)\big)$$



$\phi(t)$ is the (strict) generator function of the Archimedean copula. ϕ^{-1} is its inverse.

Suppose $\phi : [0,1] \rightarrow [0,\infty)$ is continuously strictly decreasing s.t. $\phi(1) = 0$ and $\phi(0) = \infty$

$\phi^{-1} : [0,\infty) \rightarrow [0,1]$

05 Copulas with Large Tail Dependency

The Clayton copula function is:

$$C_C(u_1, u_2, \ldots, u_n) = \left(\sum_{i=1}^{n} u_i^{-\beta} - n + 1 \right)^{-\frac{1}{\beta}}, \ \beta > 0.$$

for $n \geq 3$.

As u_i's ↑1, C ↑ 1
As u_i's ↓0, C ↓0

For the Clayton pdf,

$$f_C(u_1, u_2, \ldots, u_n) \equiv \frac{\partial^n C}{\partial u_1 \partial u_2 \ldots \partial u_n} \prod_{i=1}^{n} \frac{\partial u_i}{\partial z_i} = ABD \prod_{i=1}^{n} f_i(z_i), \qquad (18.2)$$

where $f_i(z_i)$ is the pdf of r.v. z_i, and

$$A = \left(\prod_{j=1}^{n} u_j \right)^{-(\beta+1)}, \quad B = \prod_{j=1}^{n-1} (1+j\beta), \quad D = \left[\sum_{j=1}^{n} u_j^{-\beta} - n + 1 \right]^{-\left(\frac{1}{\beta}+n\right)}.$$

Consider the tail dependency of the bivariate Clayton copula $C(u,u) = (2\,u^{-\beta} - 1)^{-1/\beta} = [2 - u^{\beta}]^{-1/\beta}\,u$.
Hence tail dependency = $\lim_{u\downarrow 0} C(u,u)/u = \lim_{u\downarrow 0} [2 - u^{\beta}]^{-1/\beta} = 1/2^{1/\beta}$ which is > 0 for $\beta > 0$.

05 Copulas with Large Tail Dependency

As u_i's ↑1, C ↑ 1
As u_i's ↓0, C ↓0

The Gumbel copula is:

$$C_G(u_1, u_2, \ldots, u_n) = \exp\left\{-\left[\sum_{i=1}^{n}(-\ln u_i)^{\beta}\right]^{\frac{1}{\beta}}\right\}, \ \beta > 1.$$

for $n \geq 3$.

For Gumbel pdf,

$$f_G(u_1, u_2, \ldots, u_n) = C_G BP \prod_{i=1}^{n} f_i(z_i), \tag{18.3}$$

where

$$B = \prod_{i=1}^{n}\left[\frac{1}{u_i}(-\ln u_i)^{\beta-1}\right], \quad P = \sum_{i=1}^{n} a_i^n A^{\frac{1}{\beta}-n},$$

$A = \sum_{i=1}^{n}(-\ln u_i)^{\beta}$, and a_i^n is defined as follows: $a_1^{k+1} = a_1^k(k\beta - 1), a_2^{k+1} = a_1^k + a_2^k(k\beta - 2)$, $a_3^{k+1} = a_2^k + a_3^k(k\beta - 3), \ldots, a_k^{k+1} = a_{k-1}^k + a_k^k(k\beta - k)$, and $a_{k+1}^{k+1} = 1$, for $1 \leq i \leq n$.

05 Copulas with Large Tail Dependency

As u_i's $\uparrow 1$, C $\uparrow$ 1
As u_i's $\downarrow 0$, C $\downarrow 0$

The Frank copula is:

$$C_F(u_1, u_2, \ldots, u_n) = -\frac{1}{\beta} \ln \left\{ 1 + \frac{\prod_{i=1}^{n} \left(e^{-\beta u_i} - 1\right)}{\left(e^{-\beta} - 1\right)^{n-1}} \right\}, \ \beta > 0$$

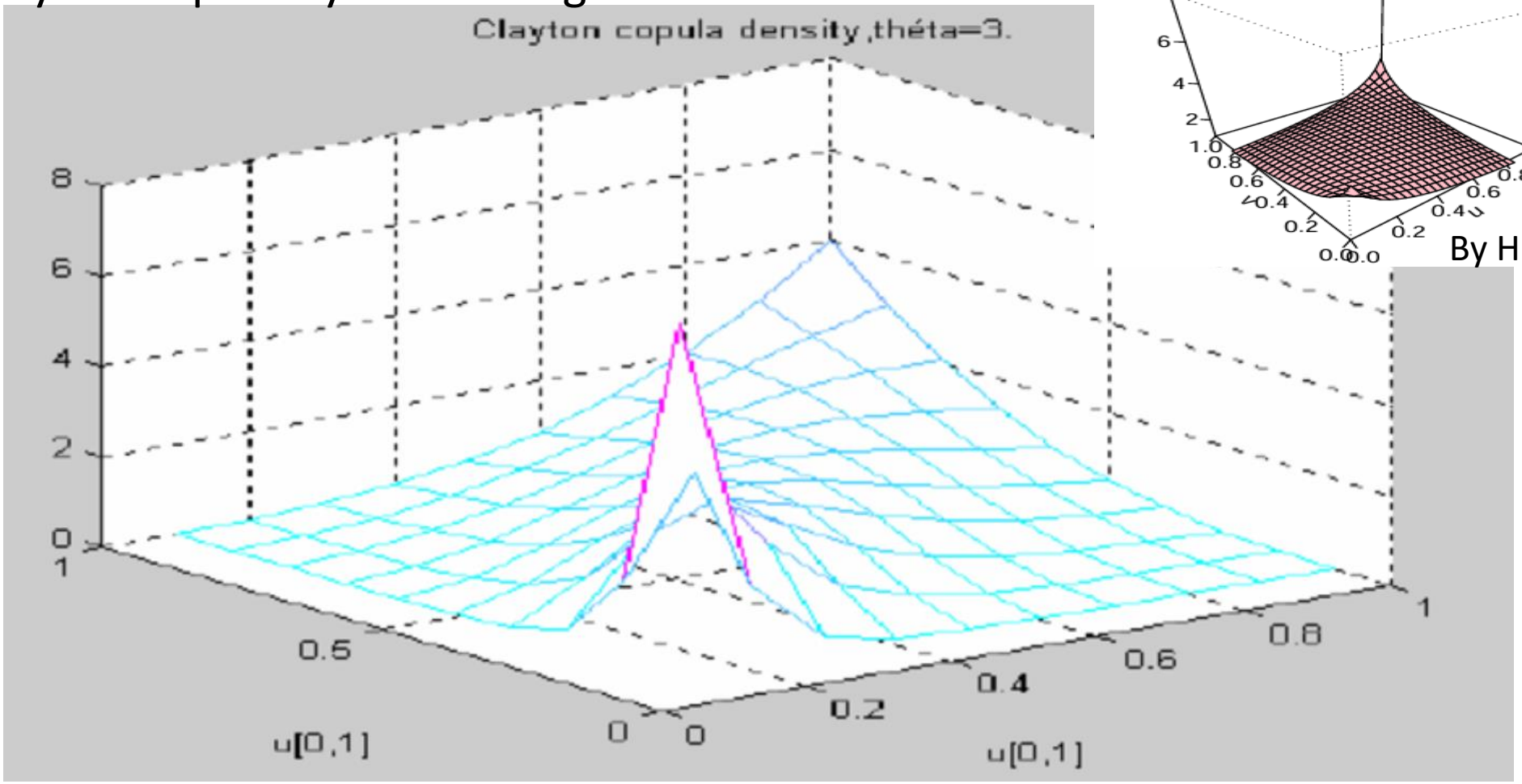
for $n \geq 3$.

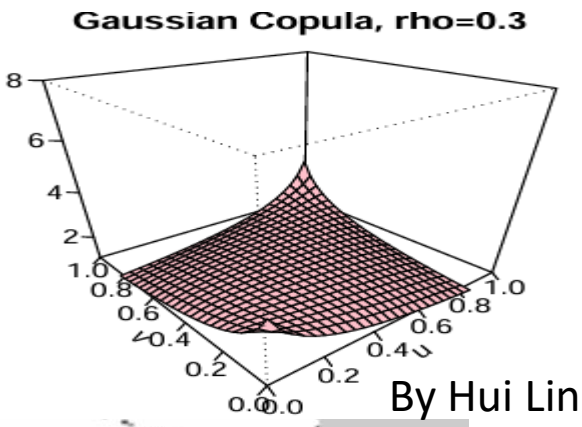
For Frank pdf,

$$f_F(u_1, u_2, \ldots, u_n) = \left(\sum_{j=1}^{n} c_j A^{-j} P_j \right) \prod_{i=1}^{n} f_i(z_i), \qquad (18.4)$$

where $A = 1 + \frac{\prod_{i=1}^{n}(e^{-\beta u_i}-1)}{(e^{-\beta}-1)^{n-1}}$, $P_j = \frac{(-\beta)^{n-1} \prod_{k=1}^{n} e^{-\beta u_k} [\prod_{k=1}^{n}(e^{-\beta u_k}-1)]^{j-1}}{(e^{-\beta}-1)^{j(n-1)}}$, $c_j = (-1)^{j-1}(j-1)!a_j$, and $a_j \equiv S(n,j)$ is the Stirling number of the second kind where $S(n,j) = \frac{1}{j!} \sum_{k=0}^{j} (-1)^{j-k} \binom{j}{k} k^n$.

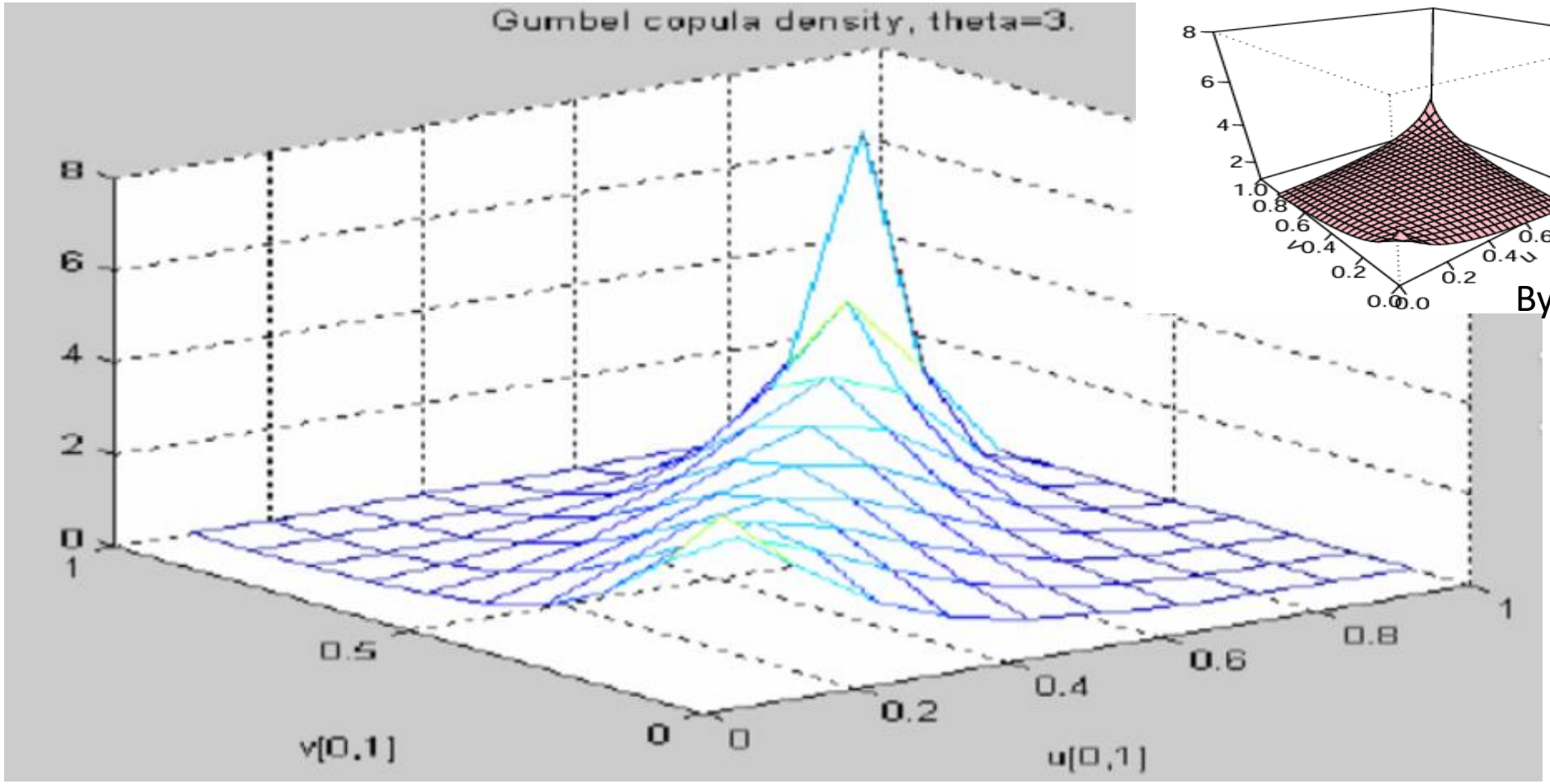
Illustration of High Lower End Tail Dependencies in Clayton Copula by Research-gate





By Hui Lin

Illustration of High Higher End Tail Dependencies in Gumbel Copula by Research-gate



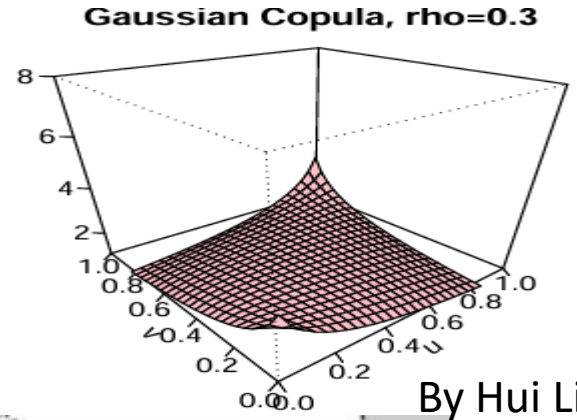


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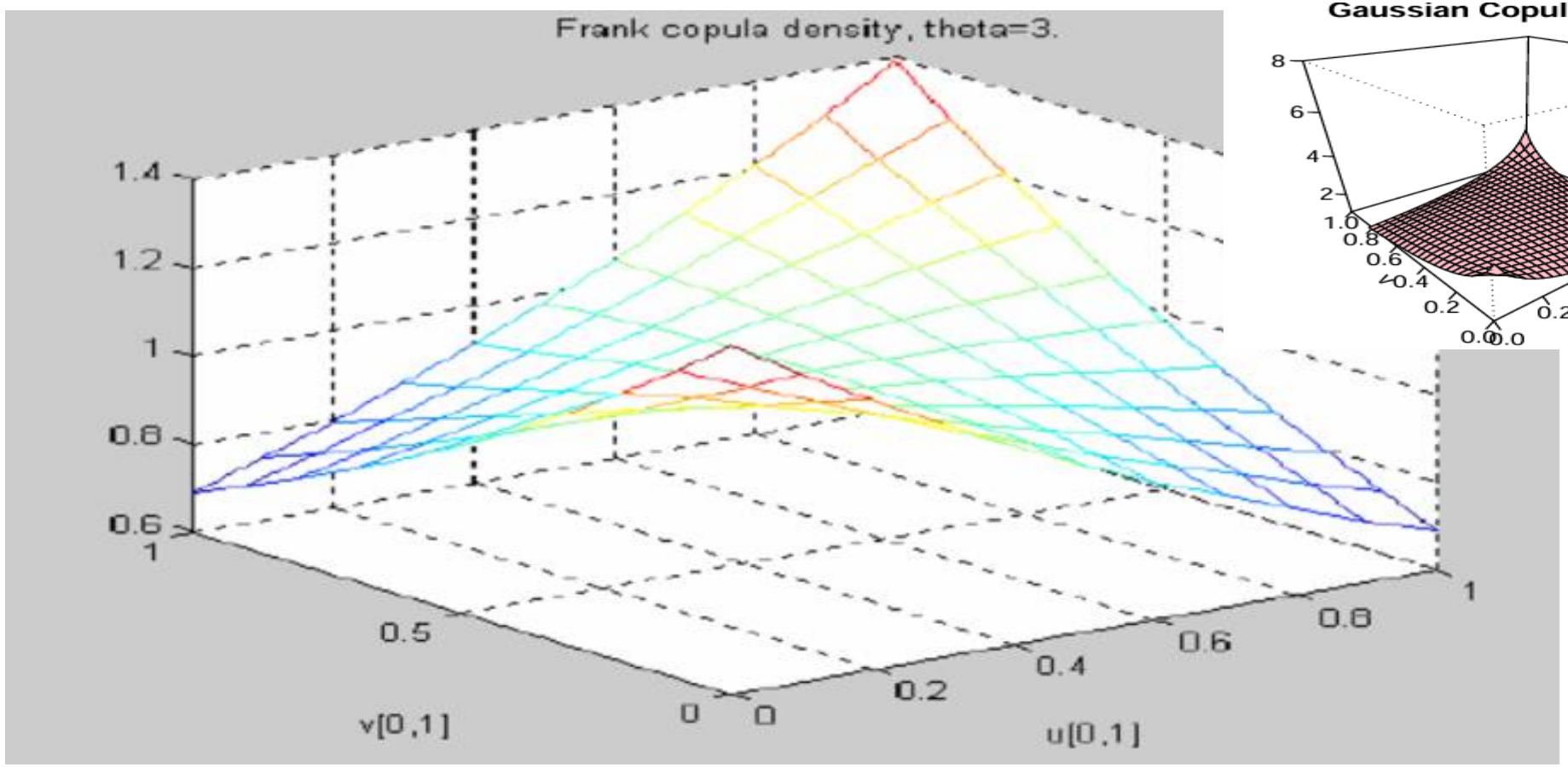
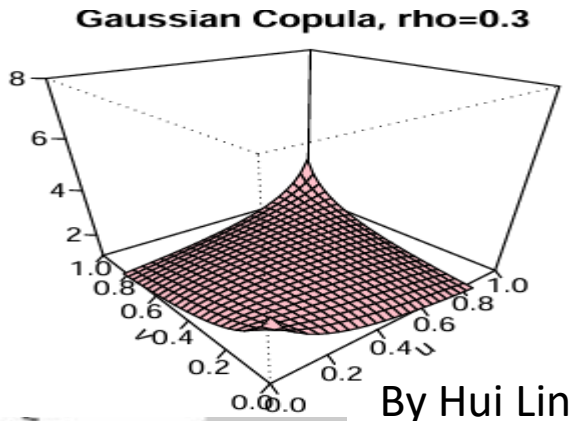
Illustration of Slightly Higher Low & High End Tail Dependencies in Frank Copula by Research-gate





By Hui Lin

06 Estimating Copulas

There are several related methods that can be used for estimating copulas:

1. Maximum likelihood estimation (MLE). It is often considered too difficult to apply as there too many parameters to estimate.

2. Pseudo-MLE of which there are two types: *parametric* pseudo-MLE and *semi-parametric* pseudo-MLE. Pseudo-MLE seems to be used most often in practice. The marginals are estimated via their *empirical CDFs* and the copula is then estimated via MLE.

3. Moment-matching methods are also sometimes used. They methods can also be used for finding starting points for (pseudo)-MLE.

Maximum Likelihood Estimation

Let $\mathbf{Y} = (Y_1 \dots Y_d)^\top$ be a random vector and suppose we have parametric models $F_{Y_1}(\cdot \mid \boldsymbol{\theta}_1), \dots, F_{Y_d}(\cdot \mid \boldsymbol{\theta}_d)$ for the marginal CDFs. We assume that we also have a parametric model $c_{\mathbf{Y}}(\cdot \mid \boldsymbol{\theta}_C)$ for the copula density of $\mathbf{Y}$. By differentiating (2) we see that the density of $\mathbf{Y}$ is given by

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{Y}}(y_1, \dots, y_d) = c_{\mathbf{Y}}\left(F_{Y_1}(y_1), \dots, F_{Y_d}(y_d)\right) \prod_{j=1}^{d} f_{Y_j}(y_j). \tag{14}$$

06 Estimating Copulas

Maximum Likelihood Estimation

Suppose now that we are given an IID sample $\mathbf{Y}_{1:n} = (\mathbf{Y}_1, \ldots, \mathbf{Y}_n)$. We then obtain the log-likelihood as

$$\begin{aligned}
\log L(\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_d, \boldsymbol{\theta}_C) &= \log \prod_{i=1}^{n} f_{\mathbf{Y}}(\mathbf{y}_i) \\
&= \sum_{i=1}^{n} \Big(\log[c_{\mathbf{Y}}\left(F_{Y_1}(y_{i,1} \,|\, \boldsymbol{\theta}_1), \ldots, F_{Y_d}(y_{i,d} \,|\, \boldsymbol{\theta}_d) \,|\, \boldsymbol{\theta}_C\right)] \\
&\quad + \log\left(f_{Y_1}(y_{i,1} \,|\, \boldsymbol{\theta}_1)\right) + \cdots + \log\left(f_{Y_d}(y_{i,d} \,|\, \boldsymbol{\theta}_d)\right) \Big). \qquad (15)
\end{aligned}$$

The ML estimators $\widehat{\boldsymbol{\theta}}_1, \ldots, \widehat{\boldsymbol{\theta}}_d, \widehat{\boldsymbol{\theta}}_C$ are obtained by maximizing (15) with respect to $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_d, \boldsymbol{\theta}_C$. There are problems with this approach, however:

1. There are (tto) many parameters to estimate, especially for large values of d. As a result, performing the optimization can be difficult.
2. If any of the parametric univariate distributions $F_{Y_i}(\cdot \,|\, \boldsymbol{\theta}_i)$ are misspecified then this can cause biases in the estimation of both the univariate distributions and the copula.

06 Estimating Copulas

Pseudo-Maximum Likelihood Estimation

The pseudo-MLE approach helps to resolve the problems associated with the MLE approach mentioned above. It has two steps:

1. First estimate the marginal CDFs to obtain $\widehat{F}_{Y_j}$ for $j = 1, \ldots, d$. We can do this using either:
   - The *empirical CDF* of $y_{1,j}, \ldots, y_{n,j}$ so that
   $$\widehat{F}_{Y_j}(y) = \frac{\sum_{i=1}^{n} 1_{\{y_{i,j} \leq y\}}}{n+1}.$$
   - A parametric model with $\widehat{\theta}_j$ obtained using the usual univariate MLE approach.
2. Then estimate the copula parameters $\boldsymbol{\theta}_C$ by maximizing
$$\sum_{i=1}^{n} \log \left[c_{\mathbf{Y}} \left(\widehat{F}_{Y_1}(y_{i,1}), \ldots, \widehat{F}_{Y_d}(y_{i,d}) \,|\, \boldsymbol{\theta}_C \right) \right] \tag{16}$$

Note that (16) is obtained directly from (15) by only including terms that depend on the (as yet) unestimated parameter vector $\boldsymbol{\theta}_C$ and setting the marginals at their fitted values obtained from Step 1.

06 Estimating Copulas

Fitting Gaussian and t Copulas (matching Kendall's tau with Pearson's rho)

- Y is a meta-Gaussian probability distribution if Y = Ψ(X), and X is normally distributed. Ψ(.) is an increasing continuous function.

A moment-matching approach for fitting either the Gaussian or t copulas can be obtained immediately from the following result.

Proposition 7
Let $\mathbf{Y} = (Y_1 \ldots Y_d)^\top$ *have a meta-Gaussian distribution with continuous univariate marginal distributions and copula* $C_{\mathbf{\Omega}}^{Gauss}$ *and let* $\Omega_{i,j} = [\mathbf{\Omega}]_{i,j}$. *Then*

$$\rho_\tau(Y_i, Y_j) = \frac{2}{\pi} \arcsin(\Omega_{i,j}) \tag{17}$$

- Student t-distribution has the same Kendall's tau function as Gaussian distribution.
- For Student t-distribution (v, ρ) with additional parameter degrees of freedom v, match sampling variance with v/(v-2), v>2, to find v after finding estimate of ρ.

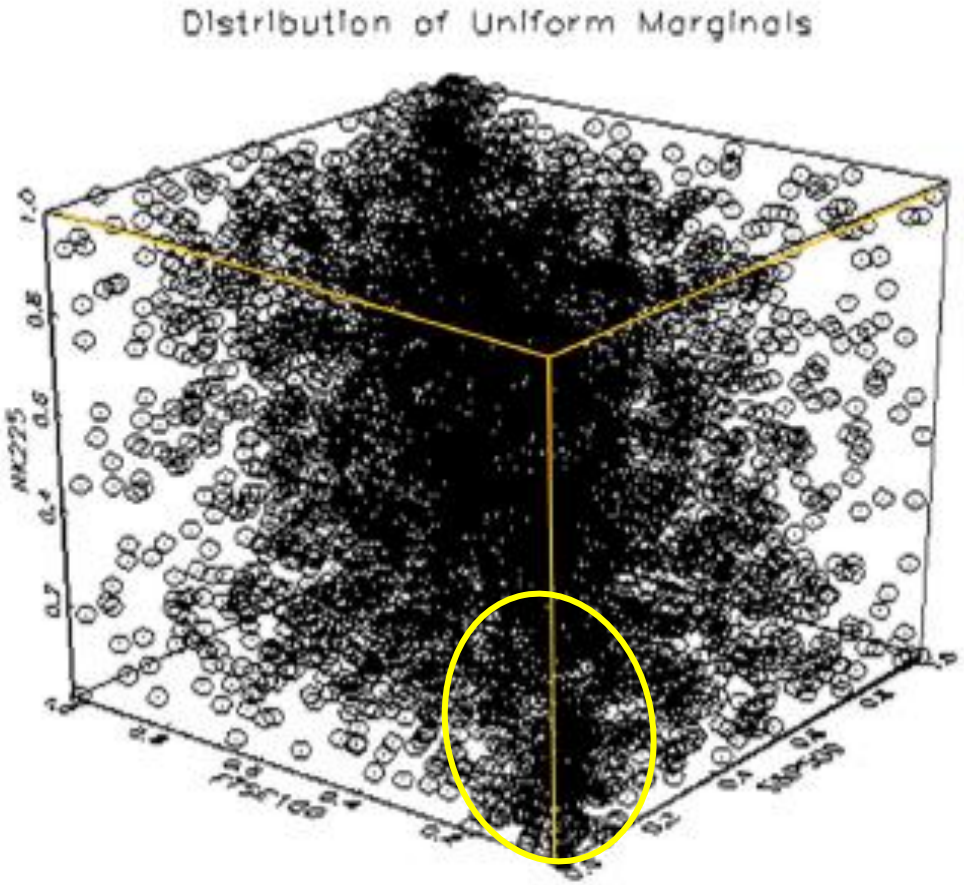
06 Estimating Copulas – Systemic Risk

Figure 18.1 shows the CDF's of the S&P500, FTSE100, and N225 Index daily returns in the sample period from January 1990 to April 2012. It shows that during the entire sampling period, the corresponding cdf uniform variables of the returns were mostly clustered around the centre of the unit cube and toward both the upper back and lower front ends.

Suppose each time series of daily returns is of sample size T. The cdf uniform variable of a return is obtained by rst sorting the T returns from the smallest to the largest value, and then assigning k=T if a return is the k_{th} value from the smallest value.

In terms of the implications to financial investments and

bank portfolios, **the lower end closest to the viewer** is the simultaneous loss region that is of particular importance and special interest in the study of contagion and systemic risk.



06 Estimating Copulas

The empirical pdf is simply one divided by the total number of sample points. To find the standard deviation of the ML estimates, we employ the jackknife method. This is useful for small samples as well as for the cases where parametric simulation is not possible due to the unknown underlying distribution. The "z-values" are computed by dividing the $\hat{\beta}$ estimates by their standard deviations, and provide an indication of whether the estimates are significantly different from zeros. Sometimes, ML based on a null empirical distribution, or the semi-parametric approach, is also called a pseudo-ML where part or one of the parameters is estimated using an alternative consistent, asymptotic method instead of full ML due to reasons such as non-convergence or having an empirical distribution as marginal. The $\hat{\beta}$ estimated in this semi-parametric approach are consistent. The semi-parametric estimation results are shown in Table 18.1.

Kullback–Leibler (KL) divergence or distance (also called **relative entropy**) measures how much a model probability distribution Q deviates from true probability distribution P.

$$\sum_x P(x)\, ln\frac{P(X)}{Q(x)} \geq 0$$

The KL divergence of *P* from *Q* may be interpreted as the expected excess surprise from using Q as the model instead when the actual distribution is *P*.

The estimated Clayton copula is seen to have the minimum KL distance and largest maximum likelihood.

Table 18.1. Maximum Likelihood Estimates and Kullback-Leibler Distances.

Copula	Estimate $\hat{\beta}$	Jackknife Std. dev.	"z-value"	Kullback-Leibler Distance	ML
Student-t	6	0	∞	5.68	-10,251
Clayton	1.15	0.017	67.6	3.72	-11.92
Gumbel	2.12	0.034	62.3	4.10	-1,982
Frank	4.29	0.052	82.4	4.29	-2,998

		Copula	beta	C(u1,u2,u3)
u1	0.1	Clayton	1.15	0.040121
u2	0.1	Gumbel	2.12	0.020940
u3	0.1	Frank	4.29	0.010401
		Gaussian	Σ	0.000983

$$\Sigma = \begin{bmatrix} 1 & 0.0083 & -0.0344 \\ 0.0083 & 1 & 0.0202 \\ -0.0344 & 0.0202 & 1 \end{bmatrix}$$

06 Estimating Copulas – An Application in CDO

Suppose there is a simple one-period Credit Default Obligation (CDO) with the following characteristics. The maturity is 1 year. There are N = 125 bonds in the reference portfolio. Each bond pays a coupon of one unit after 1 year if it has not defaulted. The recovery rate on each defaulted bond is zero. There are 3 tranches of interest: the equity, mezzanine and senior tranches with attachment points 0-3 defaults, 4-6 defaults and 7-125 defaults, respectively.

We make the simple assumption that the probability, q, of defaulting within 1 year is identical across all bonds X_i is the normalized asset value of the ith credit, i.e. bond, and we assume

$$X_i = \sqrt{\rho} M + \sqrt{1-\rho}\, Z_i \tag{19}$$

where $M, Z_1, \ldots, Z_N$ are IID normal random variables. Note that the correlation between each pair of asset values is identical. We assume also that the ith credit defaults if $X_i \leq \bar{x}_i$. Since the probability of default, q, is identical across all bonds we must therefore have

$$\bar{x}_1 = \cdots \bar{x}_N = \Phi^{-1}(q). \tag{20}$$

It now follows from (19) and (20) that

Note: q is risk-netural probability of default implied from some bond pricing.

06 Estimating Copulas – An Application in CDO

$$\begin{aligned}
\mathsf{P}(\text{Credit } i \text{ defaults} \,|\, M) &= \mathsf{P}(X_i \le \bar{x}_i \,|\, M) \\
&= \mathsf{P}(\sqrt{\rho}M + \sqrt{1-\rho}\, Z_i \le \Phi^{-1}(q) \,|\, M) \\
&= \mathsf{P}\left(Z_i \le \frac{\Phi^{-1}(q) - \sqrt{\rho}M}{\sqrt{1-\rho}} \,\middle|\, M \right).
\end{aligned}$$

Therefore conditional on M, the total number of defaults is Binomial(N, q_M) where

$$q_M := \Phi\left(\frac{\Phi^{-1}(q) - \sqrt{\rho}M}{\sqrt{1-\rho}} \right).$$

That is,

$$p(k \,|\, M) = \binom{N}{k} q_M^k (1 - q_M)^{N-k}. \tag{21}$$

The unconditional probabilities can be computed by integrating numerically the binomial probabilities with respect to M so that

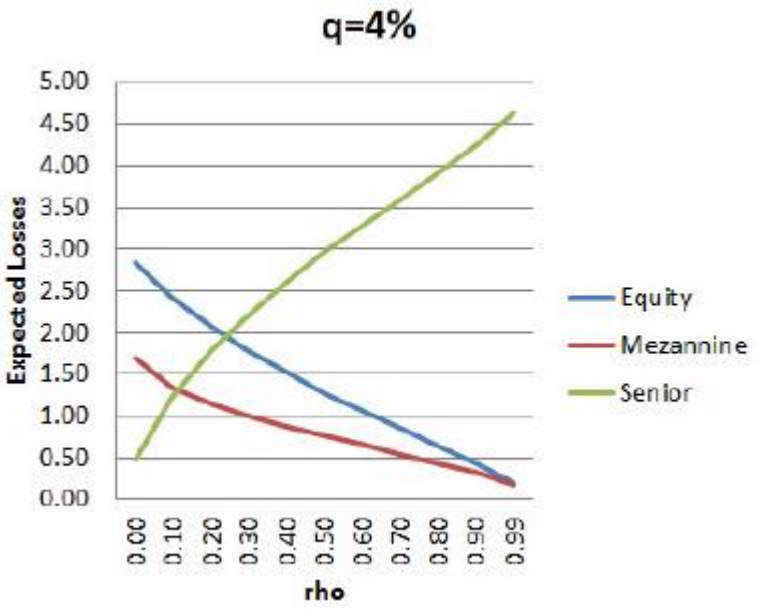
$$\mathsf{P}(k \text{ defaults}) = \int_{-\infty}^{\infty} p(k \,|\, M)\phi(M)\, dM \tag{22}$$

where $\phi(\cdot)$ is the standard normal PDF.

06 Estimating Copulas – An Application in CDO

The expected (risk-neutral) losses on each of the three tranches are:

$$\begin{aligned}
\mathrm{E}_0^Q\,[\text{Equity tranche loss}] &= 3 \times \mathrm{P}(3 \text{ or more defaults}) + \sum_{k=1}^{2} k\,\mathrm{P}(k \text{ defaults}) \\
\mathrm{E}_0^Q\,[\text{Mezzanine tranche loss}] &= 3 \times \mathrm{P}(6 \text{ or more defaults}) + \sum_{k=1}^{2} k\,\mathrm{P}(k+3 \text{ defaults}) \\
\mathrm{E}_0^Q\,[\text{Senior tranche loss}] &= \sum_{k=1}^{119} k\,\mathrm{P}(k+6 \text{ defaults}).
\end{aligned}$$



Expected Tranche Losses As a Function of q and

Homework 3 (10%)

Complete the 5 Questions in Elearn Part 2 – Homework 3. Submit answers to Elearn by the start of Lecture 9.